

List of Practical's

Week	S. No.	Experiment / Problem Statement	Contact Hours
Week 1	1	Design, develop and implement a C program to implement Euclid's Algorithm for finding the Greatest Common Divisor (GCD) of two numbers using recursion.	2
	2	Design, develop and implement a C program to solve the Tower of Hanoi problem using recursion.	
Week 2	3	Design, develop and implement a C program to generate all permutations of a given set of elements using recursion.	2
	4	Design, develop and implement a C program to verify recurrence relations using the Iteration (Substitution) Method and compare theoretical complexity with observed results.	
Week 3	5	Design, develop and implement a C program to implement Binary Search using the Divide and Conquer technique.	2
	6	Design, develop and implement a C program to implement Merge Sort and analyze its execution time.	
Week 4	7	Design, develop and implement a C program to implement Quick Sort and analyze its execution time.	2
	8	Design, develop and implement a C program to implement Randomized Quick Sort and compare its performance with Quick Sort.	
Week 5	9	Design, develop and implement a C program to find the Maximum and Minimum elements in an array using the Divide and Conquer approach.	2
	10	Design, develop and implement a C program to find the Kth Smallest Element using a Selection Algorithm.	
Week 6	11	Design, develop and implement a C program to implement Counting Sort.	2
	12	Design, develop and implement a C program to implement Radix Sort.	
Week 7	13	Design, develop and implement a C program to solve the Fractional Knapsack Problem using the Greedy approach.	2
	14	Design, develop and implement a C program to implement Job Sequencing with Deadlines using the Greedy method.	
Week 8	15	Design, develop and implement a C program to solve the Activity Selection Problem using the Greedy approach.	2
	16	Design, develop and implement a C program to find the Minimum Cost Spanning Tree using Prim's Algorithm.	
Week 9	17	Design, develop and implement a C program to find the Minimum Cost Spanning Tree using Kruskal's Algorithm.	2
	18	Design, develop and implement a C program to perform Topological Sorting of a Directed Acyclic Graph (DAG).	
Week 10	19	Design, develop and implement a C program to implement Dijkstra's Single Source Shortest Path Algorithm.	2
	20	Design, develop and implement a C program to implement the 0/1 Knapsack Problem using Dynamic Programming.	
Week 11	21	Design, develop and implement a C program to find the Longest Common Subsequence (LCS) of two strings using Dynamic Programming.	2
	22	Design, develop and implement a C program to solve the Matrix Chain Multiplication problem using Dynamic Programming.	

Week 12	23	Design, develop and implement a C program to find the All-Pairs Shortest Paths using the Floyd–Warshall Algorithm.	2
	24	Design, develop and implement a C program to construct an Optimal Binary Search Tree using Dynamic Programming.	
Week 13	25	Design, develop and implement a C program to implement the Naïve String Matching Algorithm.	2
	26	Design, develop and implement a C program to implement the Knuth–Morris–Pratt (KMP) String Matching Algorithm and compare its performance with the Naïve approach.	
Week 14	27	Design, develop and implement a C program to implement the Rabin–Karp String Matching Algorithm.	2
	28	Design, develop and implement a C program to solve the Sum of Subsets Problem using the Backtracking technique.	
Week 15	29	Design, develop and implement a C program to solve the 8-Queens Problem using the Backtracking technique and analyze the search space.	2
	30	Design, develop and implement a C program to solve any one of the following using Branch and Bound: Graph Coloring, Hamiltonian Cycle and compare the solution with exhaustive search.	